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Research on dam-break induced tsunami bore acting on the triangular breakwater based on high order 3D CLSVOF-THINC/WLIC-IBM approaching

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ABSTRACT

In present study, high order three dimensional coupled level set and volume of fluid (CLSVOF) based simulations about dam-break induced tsunami bore interacting with the stationary triangular breakwater was numerically investigated. Under our CLSVOF approaching, VOF advection equation is solved by tangent of hyperbola for interface capturing (THINC) with weighed linear interface calculation (WLIC) scheme, and level set advection equation is solved directly by the high-resolution optimized compact reconstruction weighted essentially non-oscillatory (WENO) scheme. Navier-Stokes equations is solved by a projection method on a fixed staggered Cartesian finite difference mesh. The pressure Poisson equation is solved by the point successive over-relaxation (PSOR) method. The direct forcing immersed boundary method (IBM) was implemented for breakwater-tsunami bore interface treatment. Numerical experiments based on our high order 3D CLSVOF-THINC/WLIC-IBM method were carried out in a rectangular water channel with a stationary typical triangular breakwater. The numerical results were compared with corresponding laboratory experimental results and reasonable agreements were obtained with satisfied accuracy. Numerical cases with different shape of triangular breakwater are carried out then in order to investigate the free surface evolution and energy dissipation due to the breakwater and useful conclusion is obtained.

1. Introduction

Tsunami is nearly the most destructive ocean disasters in the world, which is mainly induced by coastal and submarine landslides and oceanic earthquakes. For example, the December 2004 Indian Ocean Tsunami, the March 2010 Chile Tsunami and the March 2011 Japan Tsunami caused severe damage in recent years causes serious disasters, such as numerous losses of life and economic of coastal area underscored the need for a better understanding of tsunami wave and nearshore structure interaction (Linton. et al., 2013). When tsunami waves propagate into nearshore region, the wave energy will focus and therefore wave amplitude increases significantly due to the decrease of water depth (Kocaman and Cagatay, 2015a). As the result, to predict the arriving time and the maximum run-up height of tsunami wave and its energy dissipation caused by nearshore breakwater by means of numerical models has become an important missions of the tsunami research (Yang et al., 2018a).

Currently, as the most commonly used two methods of tsunami wave simulation in physical or numerical model experiments, isolated wave and dam-break wave are widely used to investigate the interaction of tsunami waves with various nearshore structures, Since dam-break induced wave propagation can be considered as wave propagation caused by a tsunami wave breaking on the coastline that was proposed by (Visscher and Hager, 1998; Chanson et al., 2000). Especially, the dam-break induced wave is capable to better appear the tsunami bore behaviors in the water flume, hence physical and numerical models of dam-break flows induced tsunami bore have received considerable attention in the past few years (Nouri. et al., 2010; Al-Faesly. et al., 2012; Wei. et al., 2014).

A free surface of dam-break flow is usually subject to transient large and complex deformations. An efficient and useful numerical scheme should able to both accurately predict the free surface position and maintain mass conservation in the same time (Lee and Sheu, 2001). Shallow water equations had been widely adopted in studying dam-break flow as the traditional way (Aureli. et al., 2008). Shallow water equations is the suitable approaching way to describe dam-break flows

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due to its less computational cost (Hu et al., 2015; Brufau. and Garcia-Navarro., 2000). However, because of its 2-D assumptions, Shallow water equations are inadequate to accurately simulate flows especially involving significant variations of water depth, velocity, and pressure in the vertical direction, such as dam-break flow near in-stream structures. Hence, the emphasis of numerical dam-break research has shifted to the development of Navier−Stokes ($N-S$) equations based model (Park et al., 2012; Celis et al., 2017; Yang et al., 2018b). The SWEs and RANS with the $\kappa-\varepsilon$ turbulence model are compared with each other for dam-break flow impact on a vertical wall (Kocaman and Cagatay, 2015b), and their result shown that the RANS results was better than that of SWEs results comparing with experimental results. Normally, $N-S$ equations based numerical model can be divided into two different category: mesh-free model and mesh-based model. Mesh-free models, such as moving particle semi-implicit (MPS) method (Zhang and Wan, 2014; Ng et al., 2015) and smoothed particle hydrodynamics (SPH) (Kao and Chang, 2011, 2012), within that computing and capturing free surface are always relatively easy jobs, but they still requires a great number of computing resources. Two main mesh-based models to handle free surface deformation are the volume of fluid (VOF) (Hirt. and Nichols., 1981) and level set (LS) (Osher and Fedkiw, 2001, 2003) methods, respectively.

VOF method was paid much attention in the field of the hydrodynamics in these year that was widely adopted in free surface numerical modeling, due to its advantage of mass conservation. VOF methods can be categorized to both a geometrical and an algebraic methods for interface capturing. Geometrical VOF methods which involve geometrical reconstruction such as PLIC (piecewise linear interface calculation) (Parker and Youngs, 1992) are relatively complicated for coding since it use a linear approximation with the orientation of the interface considered after advection. Different to geometrical VOF method, the algebraic VOF method has been proposed and developed usefully. Typical improvement like tangent of hyperbola for interface capturing (THINC) scheme was proposed by Xiao et al. (Xiao. et al., 2005), which makes use of the hyperbolic tangent function to compute the numerical flux for the fluid fraction function, that is able to gives a conservative, oscillation-less results of the fluid fraction function, even for the extremely distorted interfaces.

Use the simple line interface calculation (SLIC) to reconstruct multi-dimensional numerical flux approximations and meantime consider the orientation of the interface have been reported in (Yokoi., 2007), which is called VOF and weighed line interface calculation (VOF/WLIC) scheme. An interface orientation dependent weighting between the original THINC scheme (Xiao. et al., 2011) and the first order donor cell scheme is suggested in (Yokoi., 2007), which greatly enhance the geometrical accuracy of the original THINC scheme. THINC/SW (THINC with Slope Weighting) scheme was proposed by Xiao et al. (Xiao. et al., 2011) that presents an improved multi-dimensional algebraic VOF method to capture moving interfaces. According to the interface orientation, the interface jump in the THINC scheme is adaptively scaled to a proper thickness. Application of dam-break problems by VOF method were carried out to investigates its nonlinear flow behaviors (Liu. et al., 2013; Marsooli and Wu, 2014).

The LS based method has also become more and more popular for simulating free surface between gas and liquid, due to its advantages that the interface is defined by the zero contour of a continuous signed distance function (Yang et al., 2014; Zhu et al., 2016). Such approaching allows the normality and curvature of the interface are easily obtained. Plus, by means of implementing a high-order advection scheme, the complex topology changes is also easy to deal with (Kim and Liou, 2011). But the typical LS based methods contains two important issues. First issue is that the deterioration due to numerical dissipation and dispersion in some occasion. A useful suggestion is to adopt high-order schemes such as weight essentially non-oscillatory (WENO) based schemes to handle the advection process for the LS equation (Hermann, 2014; Salih and Moulic, 2009). Second issue is that mass conservation. A small amount of mass may be gained or lost when the zero level set is altered during a re-initialization procedure (Luo et al., 2015). A re-initialization procedure which was adopted in Sussman's research is suggested for improving the mass conservation (Sussman and Fatemi, 1999).

An effective option and reasonable technique is to adopt a hybrid approach including LS and VOF, such as coupled LS/VOF (CLSVOF) method (Wang and Tong, 2010; Wang et al., 2012; Balczar et al., 2016). CLSVOF method gathers the advantages from both VOF and LS, that means it able to further improve mass conservation and calculate curvature of the interface accurately from VOF and LS methods respectively. To approach such hybrid procedure is first using the VOF function to advect the free surface, then use the smoothed LS function to obtain the geometrical deformation of the free surface. The improved re-initialization procedure for LS equation was also usually adopted within CLSVOF approaching (Chakraborty et al., 2013; Ningegowda and Premachandran, 2014Li and Yu, 2019). This coupling method was widely used for high precision simulation such as droplet impacting a dry surface (Yokoi et al., 2009).

The triangular breakwater, as a typical nearshore structure, has been widely applied to dissipate the wave energy and protect the people and properties behind it (Hatice et al., 2014). It is always complex and difficult to change the slope of the model in the laboratory experiment (Cheng et al., 2017). But the immersed boundary method (IBM) is able to simulate the breakwater-tsunami bore interface involving stationary or moving bodies which are geometrically complex. IBM is applicable to problems with irregular solid objects and does not necessarily have to conform to Cartesian grids. Since the grid does not necessarily conform to the solid boundary, the governing equations near the boundary should be modified by imposing a proper boundary condition. According to the review paper of Mittal and Iaccarino (2005) the IBM can be broadly categorized into two classes, the continuous forcing and the discrete forcing. In the continuous forcing method, an explicit forcing term is introduced to the continuous $N-S$ equations before discretization. Typical examples can be seen in the original method of Peskin (1972). In comparison with the continuous forcing IBM, a sharper representation of the immersed boundary is allowed in the discrete forcing methods (Ghias et al., 2007). Hence, the IBM is powerful to apply into ocean and coastal engineering region, since the geometry and the shape of breakwater are relatively easy to model by means of IBM (Sassa et al., 2016; Christensen et al., 2018; Mondal and Alam, 2018).

In the present study, we developed high accuracy 3D numerical model based on CLSVOF combine THINC/WLIC scheme with IBM, to investigate the dam-break induced tsunami bore acting on the triangular breakwater. The main purpose of this research is to develop a 3D numerical solver which is able to accurately and efficiently predict dam break induced tsunami wave evolution and its interaction with breakwater with different shapes. This paper is organized as follows: In Section 2, a whole numerical algorithm is given, which makes sure that LS function is correctly connected to the identical geometry described by VOF function. Section 3 is presented to describe the interface on the splitting algorithm for CLSVOF equation. Advection of VOF is solved under the framework of THINC/WLIC, and LS is solved directly from its advection equation by the high-resolution OCRWENO scheme. The IBM is implemented for the evaluation of momentum forcing term, which adopts the volume fraction of solid on the body surface or inside the body. In section 4, numerical experiments were carried out first in a rectangular flume with one triangular breakwater which was setted as the benchmark case, to carry out mesh convengence analysis and mass conservation analysis, then compared with the model experiment in order to demonstrate their resolution. Based on that, in section 5 numerical investigate for tsunami bore towards the breakwater, with different slopes was carried out and discussed. The conclusions are written in Section 6.

2. Mathematical model

2.1. Level set (LS) method

In LS method, the interface can be tracked by solving the LS evolution equation as follow

$$\phi_t + \mathbf{u}\cdot\nabla\phi = 0, \tag{1}$$

where ϕ is the level set function normally setted as a signed distance function $\bar{d}$, from the space point to the interface. $\bar{d}$ is defined to be positive for liquid, negative for gas. The interface level always located at zero ψ such as

$$\phi(x, t=0) = \begin{cases} \bar{d}; & \text{for } x, y, z \text{ in the liquid} \\ 0; & \text{for } x \in \psi \\ -\bar{d}; & \text{for } x, y, z \text{ in the gas,} \end{cases} \tag{2}$$

A re-initialization equation is solved to re-keep ϕ as a signed distance function as follow

$$\phi_\tau = \bar{S}(\phi_0)(1 - |\nabla\phi|) + \lambda\delta(\phi)|\nabla\phi|, \tag{3}$$

where τ is the pseudo-time for the iteration and ϕ_0 is the level set function before re-initialization. λ and delta function $\delta(\phi)$ are respectively given as

$$\lambda = -\frac{\int_\Omega \delta(\phi)(\bar{S}(\phi_0)(1 - |\nabla\phi|))d\Omega}{\int_\Omega \delta^2(\phi)|\nabla\phi|d\Omega}, \tag{4}$$

$$\delta(\phi) = \begin{cases} \frac{1}{2\varepsilon}\left(1 + \cos\left(\frac{\pi\phi}{\varepsilon}\right)\right); & \text{if } |\phi| \le \varepsilon \\ 0; & \text{otherwise,} \end{cases} \tag{5}$$

Note that $\bar{S}(\phi_0)$ denotes the smoothed sign function.

2.2. Volume of fluid (VOF) method

The characteristic function χ is adopted in VOF method with following form

$$\frac{\partial\chi}{\partial t} + \nabla\cdot(\mathbf{u}\chi) - \chi\nabla\cdot\mathbf{u} = 0. \tag{6}$$

with

$$\chi(x, y, z) = \begin{cases} 1; & \textit{for } (x, y, z) \in \textit{the liquid} \\ -1; & \textit{for } (x, y, z) \in \textit{the gas,} \end{cases} \tag{7}$$

As an integral of χ, the fraction function C is used

$$C_{i,j,k} = \frac{1}{\Delta x\Delta y\Delta z}\int_0^{\Delta x}\int_0^{\Delta y}\int_0^{\Delta z}\chi(x, y, z)dxdydz, \tag{8}$$

2.3. Coupled LS and VOF (CLSVOF) method

The CLSVOF method takes advantage of the mass conservativeness of the VOF method and the interface smoothness of the LS method. In the CLSVOF method, a new LS field ϕ is introduced where the interface position is defined by the iso-line $\phi = 0$. First defined the volume fraction F, in a grid cell Ω of the domain, at time t, as a function of the LS ϕ:

$$F(\Omega, t) = \frac{1}{|\Omega|}\int_\Omega H(\phi(x, y, z, t))dxdy, \tag{9}$$

where H is the Heaviside function:

$$H(\phi) = \begin{cases} 1; & \text{if } \phi > 0 \\ 0; & \text{otherwise,} \end{cases} \tag{10}$$

The CLSVOF algorithm is shown as follows: For example, if we have fraction function C^n of the VOF at time n, then we have intermediate function ψ as follows

$$\psi = 2C^n - 1. \tag{11}$$

Then, solve the following re-initialization equation of the intermediate function ψ to a steady state

$$\frac{\partial\psi}{\partial\tau} + sgn(\psi_0)(|\nabla\psi| - 1) = 0. \tag{12}$$

If $|\phi^n| \le \varepsilon$, then let

$$\phi^n = \psi \tag{13}$$

Then, solve the following improved re-initialization equation of the intermediate function ϕ to a steady state

$$\frac{\partial\phi}{\partial\tau} + sgn(\phi_0)(|\nabla\phi| - 1) - \lambda\delta(\phi)|\nabla\phi| = 0 \tag{14}$$

Finally, we have re-constructed ϕ^n.

2.4. Navier–Stokes equations and physical properties

The two immiscible (e.g. liquid and gas) incompressible viscous flows are governed by the Navier-Stokes equations

$$\nabla\cdot\mathbf{u} = 0, \tag{15}$$

$$\frac{\partial\mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u} = -\frac{1}{\rho}\nabla p + \frac{1}{\rho}\nabla\cdot[2\mu\mathbf{D}] + \mathbf{g}, \tag{16}$$

where $\mathbf{u}$ is the velocity vector, t is the time, p is the pressure, $\mathbf{D} \equiv (\nabla\mathbf{u} + (\nabla\mathbf{u})^T)$ is the deformation tensor, and $\mathbf{g}$ represents the gravitational acceleration. Each phase of density and viscosity which shown in Eq. (16) can be defined by the VOF function

$$\rho = \rho_a + (\rho_w - \rho_a)C$$
$$\mu = \mu_a + (\mu_w - \mu_a)C, \tag{17}$$

where ρ_a and ρ_w represent the air and water density, respectively.

3. Numerical scheme

3.1. LS method solver

The LS evolution equation (i.e., Eq. (1)) can be expressed as following conservation form

$$\phi_t + \nabla\cdot(\mathbf{u}\phi) = 0, \tag{18}$$

Eq. (18) leads to an ordinary differential equation (ODE) with a semi-discrete conservative finite difference scheme

$$\frac{d\phi_i}{dt} = L(u_i) = -\frac{1}{\Delta x}\left(\hat{F}_{i+\frac{1}{2}} - \hat{F}_{i-\frac{1}{2}}\right) - \frac{1}{\Delta y}\left(\hat{G}_{j+\frac{1}{2}} - \hat{G}_{j-\frac{1}{2}}\right) - \frac{1}{\Delta z}\left(\hat{H}_{k+\frac{1}{2}} - \hat{F}_{k-\frac{1}{2}}\right), \tag{19}$$

where $\hat{F}_{i+\frac{1}{2}}$, $\hat{G}_{j+\frac{1}{2}}$, $\hat{H}_{k+\frac{1}{2}}$ are the approximation of numerical flux respectively at point $x_{i+\frac{1}{2}}$, $y_{j+\frac{1}{2}}$, $z_{k+\frac{1}{2}}$, which can be reconstructed from the discrete values of $F = u\phi$, $G = v\phi$ and $H = w\phi$. The numerical fluxes at cell faces that are reconstructed using a interpolation based on the idea of a OCRWENO scheme.

As similar as evolution equation, in three dimensional the re-initialization equation (3) for the three-dimensional case can be expressed in a semi-discrete form:

$$\frac{d\phi}{dt} + H_G\left(\phi_x^+, \phi_x^-, \phi_y^+, \phi_y^-, \phi_z^+, \phi_z^-\right) = 0, \tag{20}$$

where H_G is the Godunov Hamiltonian.

The spatial discretizations such as ϕ_x^+ can be evaluated similarly by the OCRWENO scheme. The third-order total variation diminishing Runge-Kutta (TVD-RK3) is used for time advancement (Shu, 1988).

Table 1
Weighting function variants for the WLIC method.

	I	II	III
$\omega_i=$	$\frac{n_x}{\mathbf{n}}$	$1-\frac{2}{\pi}\arccos\left(\frac{n_x}{\mathbf{n}}\right)$	$\tanh\left(\frac{n_x}{\mathbf{n}}\right)$

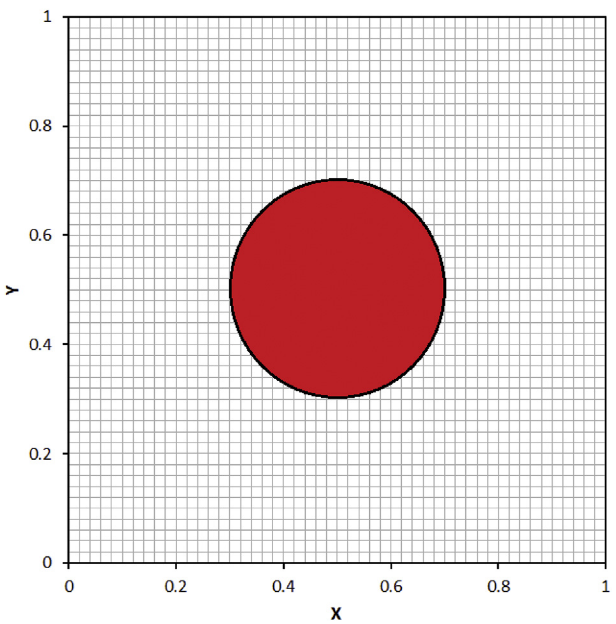


Fig. 1. The schematic diagram of validation case (unit: m).

Table 2
L_2 norm results of velocity and pressure respectively with corresponded numerical convergence order.

cell size	L_2^u	order	L_2^v	order	L_2^p	order
0.05	$4.387E^{-5}$	–	$4.164E^{-5}$	–	$7.967E^{-5}$	–
0.025	$1.142E^{-5}$	1.96	$1.062E^{-5}$	1.98	$2.117E^{-5}$	1.94
0.0125	$2.797E^{-5}$	2.02	$2.629E^{-6}$	2.01	$5.683E^{-6}$	1.93

Table 3
Wavefront arriving time on 6.15 m, 10 m and 12 m (unit: s).

	6.15 m	10 m	12 m
case 1	0.66	3.02	4.24
case 2	0.66	3.04	4.26
case 3	0.66	3.02	4.28
case 4	0.66	3.08	4.34
case 5	0.66	3.28	4.60

3.2. VOF solver

3.2.1. Weighed linear interface calculation (WLIC)

Following the idea of SLIC method, defines the horizontal interface (interface combined by x-coordinate and y-coordinate), and vertical interfaces (combined by x-coordinate and z-coordinate; combined by y-coordinate and z-coordinate), the weights of which are calculated from the surface normal $\mathbf{n}$

$$\chi_{i,j,k}(x,y,z)=\omega_{x,i,j,k}(\mathbf{n}_{i,j,k})\chi_{x,i,j,k}(x,y,z)+\omega_{y,i,j,k}(\mathbf{n}_{i,j,k})\chi_{y,i,j,k}(x,y,z) + \omega_{z,i,j,k}(\mathbf{n}_{i,j,k})\chi_{z,i,j,k}(x,y,z), \tag{21}$$

where $\mathbf{n}_{i,j,k}$ is the local normal vector and $\chi_{x,i,j,k}(x,y,z)$, $\chi_{y,i,j,k}(x,y,z)$ and $\chi_{z,i,j,k}(x,y,z)$ are the characteristic functions of the vertical interfaces and horizontal interface respectively. $\omega_{x,i,j,k}$, $\omega_{y,i,j,k}$ and $\omega_{z,i,j,k}$ in Eq. (21) are weighting functions. The choice of weighting function $\omega_{x,i,j,k}$ for x component used in Eq. (21) is arbitrary: three example possibilities are listed in Table 1 (Aniszewski et al., 2014).

The calculation of $n_{x,i,j,k}$ in Table 1 use the VOF function (Yokoi., 2007)

$$n_{x,i,j,k}=\frac{1}{4\Delta x}(C_{i+1,j,k}-C_{i,j,k}+C_{i+1,j+1,k}-C_{i,j+1,k} + C_{i+1,j,k+1}-C_{i,j,k+1}+C_{i+1,j+1,k+1}-C_{i,j+1,k+1}) \tag{22}$$

Since the spatial derivatives of VOF function are not continuous near the interface that cause geometric properties inaccurate. The calculation of $\omega_{x,i,j,k}$ presented in this study employed the signed distance function of LS method, also calculated for $\omega_{y,i,j,k}$ and $\omega_{z,i,j,k}$. For example, we calculate weighting function $\omega_{x,i,j,k}$ for x component by the following equation

$$\omega_{x,i,j,k}=\frac{|n_{x,i,j,k}|}{|n_{x,i,j,k}|+|n_{y,i,j,k}|+|n_{z,i,j,k}|}, \tag{23}$$

where $n_{x,i,j,k}=\frac{\frac{\partial\phi}{\partial x}}{\sqrt{\left(\frac{\partial\phi}{\partial x}\right)^2+\left(\frac{\partial\phi}{\partial y}\right)^2+\left(\frac{\partial\phi}{\partial z}\right)^2}}$ and $\frac{\partial\phi}{\partial x}$, $\frac{\partial\phi}{\partial y}$, $\frac{\partial\phi}{\partial z}$ are approximated by second-order centered difference scheme.

3.2.2. THINC/WLIC scheme

The flux $F_{x,x}$ is calculated based on a one-dimensional THINC scheme. As a characteristic function of the THINC scheme, the piecewise modified hyperbolic tangent function

$$\chi_{x,i}=\frac{1}{2}\left(1+\alpha_x\tanh\left(\beta\left(\frac{x-x_{i-\frac{1}{2}}}{\Delta x}-\tilde{x}_i\right)\right)\right), \tag{24}$$

is used. Here a α_x and β represent the interface direction and smoothing of the characteristic function, respectively.α_x is determined by

$$\alpha_x=\begin{cases}1; & if\ \ n_{x,i}\geq 0\\ -1; & if\ \ n_{x,i}<0,\end{cases} \tag{25}$$

Here, we used $n_{x,i}=C_{i+1}-C_{i-1}$. This $n_{x,i}$ is used only for the calculation of (24). The parameter β determines the steepness of the smoothed Heaviside function. In this paper, we use $\beta=3.5$ which corresponds one mesh spacing smoothing. $\tilde{x}_i\Delta x$ represents the distance between $x_{i-1/2}$ and the interface. $\tilde{x}_i$ calculated from the cell average of χ_i as follows

$$\frac{1}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}}\chi_i(x,\tilde{x}_i)dx=C_i^n, \tag{26}$$

We can generalize the formulation to three-dimensional case as follows

$$F_{x,i+1/2,j,k}=F_{x,i+1/2}=F_{x,x,i+1/2}+F_{x,y,i+1/2}+F_{x,z,i+1/2}, \tag{27}$$

$$F_{y,i,j+1/2,k}=F_{y,j+1/2}=F_{y,x,j+1/2}+F_{y,y,j+1/2}+F_{y,z,j+1/2}, \tag{28}$$

$$F_{z,i,j,k+1/2}=F_{z,k+1/2}=F_{z,x,k+1/2}+F_{z,y,k+1/2}+F_{z,z,k+1/2}, \tag{29}$$

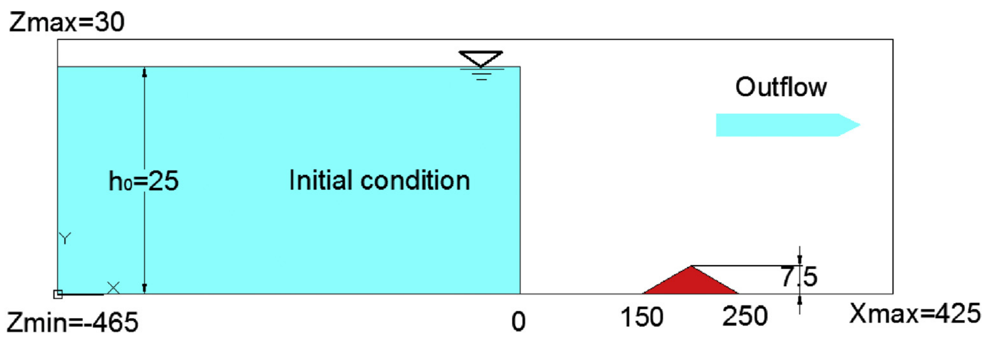
with

$$F_{\xi,\xi,m+1/2}=-\int_{\xi_{m+1/2}}^{\xi_{m+1/2}-u_{m+1/2}\Delta t}\omega_{\xi,ms}\chi_{\xi,ms}d\xi, \tag{30}$$

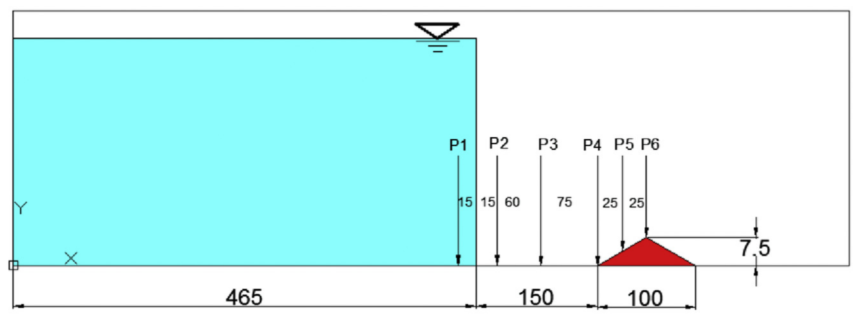
$$\chi_{\xi,ms}=\frac{1}{2}\left(1+\alpha_\xi\tanh\left(\beta\left(\frac{\xi-\xi_{ms-\frac{1}{2}}}{\Delta\xi}-\tilde{\xi}_{ms}\right)\right)\right), \tag{31}$$

$$\alpha_\xi=\begin{cases}1; & if\ \ n_{\xi,ms}\geq 0\\ -1; & if\ \ n_{\xi,ms}<0,\end{cases} \tag{32}$$

$$ms=\begin{cases}m; & if\ \ u_{\xi,m+1/2}\geq 0\\ m+1; & if\ \ u_{\xi,m+1/2}<0,\end{cases} \tag{33}$$



(a) Numerical simulation setup, lengths in (cm).



(b) Measurement positions, lengths in (cm).

Fig. 2. Numerical simulation setup for verification.

Here, the three-dimensional weights ω_{xi} are calculated as follows

$$\omega_{\xi,m} = \frac{|n_{\xi,m}|}{|n_{x,m}| + |n_{y,m}| + |n_{z,m}|}, \tag{34}$$

3.3. $N-S$ equation solver

A finite difference method is used to discretize the governing equations on a uniform staggered Cartesian grid. The velocity components u, v, and w are defined at centers of cell faces in the x, y, and z directions, respectively, while ρ, μ, p, ϕ and C are defined at cell centers.

3.3.1. Immersed boundary method (IBM)

The IBM simulates complex bodies by introducing an artificial force to the momentum equations in the vicinity of the immersed boundary Ω. The resulting momentum equations can be written as

$$\frac{\partial \underline{u}}{\partial t} + (\underline{u}\cdot\nabla)\underline{u} = -\nabla p + \frac{1}{Re}\nabla^2\underline{u} + \underline{f}, \tag{35}$$

As indicated by the name of discrete-time momentum forcing method, the forcing vector $\underline{f}$ shown in the above equation can be directly computed from the discrete-time momentum vector equation as follows

$$\frac{\underline{u}^{n+1} - \underline{u}^n}{\Delta t} + (\underline{u}\cdot\nabla)\underline{u} = -\nabla p + \frac{1}{Re}\nabla^2\underline{u} + \underline{f}, \tag{36}$$

The prescribed momentum forcing term will act only on the points nearest the immersed boundary if the boundary surface is not perfectly aligned with the grid plane. Therefore, it is required to interpolate the momentum forcing term so as to render the velocity with a magnitude approximately equal to $\underline{V}_\Omega$ at the immersed boundary Ω

$$\underline{f} = -\underline{RHS} + \frac{\underline{V}_\Omega - \underline{u}^n}{\Delta t}, \tag{37}$$

In the above equation, Δt represents the time interval and $\underline{V}_\Omega$ denotes the specified velocity along the immersed boundary. The vector $\underline{RHS}$ shown above consists of the pressure gradient, convection, and diffusion terms in the momentum equations. In the case of a stationary solid body where no-slip boundary conditions are prescribed, $\underline{V}_\Omega = 0$ will be specified along the boundary.

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} = -\frac{1}{\rho(\phi)}\frac{\partial p}{\partial x} + \frac{1}{Re}\frac{\mu(\phi)}{\rho(\phi)}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) - \frac{1}{We}\frac{\kappa(\phi)\Delta\phi\delta(\phi)}{\rho(\phi)}\frac{\partial \phi}{\partial x} + \eta F_x, \tag{38}$$

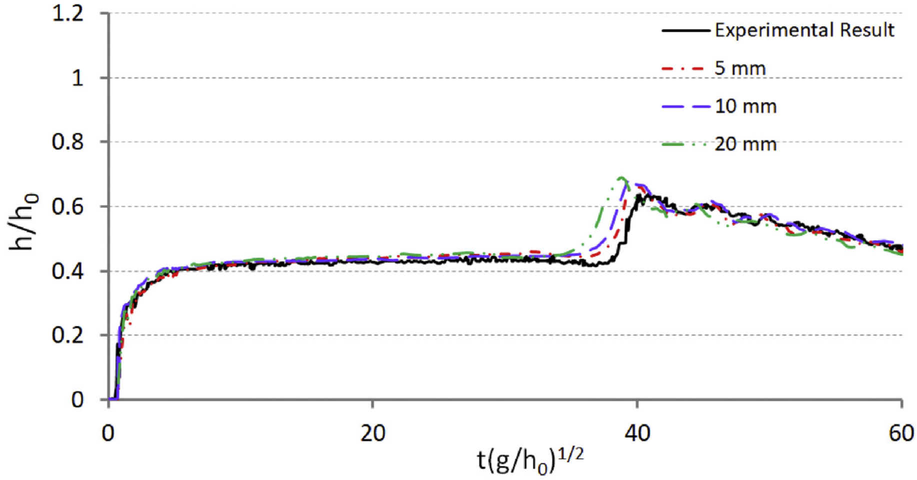
$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} = -\frac{1}{\rho(\phi)}\frac{\partial p}{\partial y} + \frac{1}{Re}\frac{\mu(\phi)}{\rho(\phi)}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right) - \frac{1}{We}\frac{\kappa(\phi)\Delta\phi\delta(\phi)}{\rho(\phi)}\frac{\partial \phi}{\partial y} + \eta F_y, \tag{39}$$

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho(\phi)}\frac{\partial p}{\partial z} + \frac{1}{Re}\frac{\mu(\phi)}{\rho(\phi)}\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right) - \frac{1}{We}\frac{\kappa(\phi)\Delta\phi\delta(\phi)}{\rho(\phi)}\frac{\partial \phi}{\partial z} - \frac{1}{Fr^2} + \eta F_z, \tag{40}$$

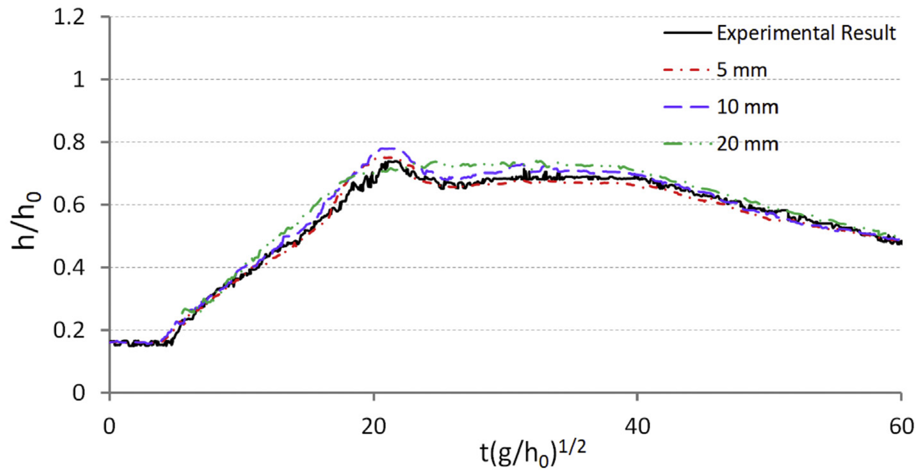
where η is the volume of fluid.

Because fluid imposes force on solid, solid can impose reacting force on fluid, so this term needs to be computed only when the grid is on the solid. In 3D grids, the center of a grid is at $\left(x_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}},\ y_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}},\ z_{i+\frac{1}{2},j+\frac{1}{2},k+\frac{1}{2}}\right)$ We can redefine η using finer grids Δx_1, Δy_1, Δz_1. Let $\Delta x/\Delta x_1 = M, \Delta y/\Delta y_1 = N, \Delta z/\Delta z_1 = L$, $M, N, L \in Z$. Therefore, the finer grid is $(x_{m,n,l}, y_{m,n,l}, z_{m,n,l})$, $m = 1, \ldots, M$, $n = 1, \ldots, N$, $l = 1, \ldots, L$, i.e.,

$$x_{m,n,l} = x_i + \frac{\Delta x_1}{2} + (m-1)\Delta x_1, \tag{41}$$



(a) P3



(b) P5

Fig. 3. Mesh convergence study results for P2 and P5.

$$y_{m,n,l} = y_j + \frac{\Delta y_1}{2} + (n-1)\Delta y_1, \tag{42}$$

$$z_{m,n,l} = z_i + \frac{\Delta z_1}{2} + (l-1)\Delta z_1, \tag{43}$$

All the points $x_{l,m,n}$, $y_{l,m,n}$, $z_{l,m,n}$ can be obtained from the above equations, and it can be seen how many points fall inside the solid. Denote the number of points falling inside the solid as NP, then the solid volume fraction can be denoted as $\eta = \frac{NP}{MNL}$ (volume of solid).

3.3.2. Momentum equation solver

The second-order upwinding scheme is used for the convective terms in Eq. (16). In the x-direction momentum equation, using $u\frac{\partial u}{\partial x}$ as an example, the discretization can be approximated as

$$u\frac{\partial u}{\partial x} = \frac{1}{2\Delta x}(u^+(3u_{i,j,k} - 4u_{i-1,j,k} + u_{i-2,j,k}) + u^-(-u_{i+2,j,k} + 4u_{i+1,j,k} - 3u_{i,j,k})), \tag{44}$$

where $u^+ = \frac{1}{2}(u_{i,j,k} + |u_{i,j,k}|)$ and $u^- = \frac{1}{2}(u_{i,j,k} - |u_{i,j,k}|)$. For the diffusion terms in Eq. (16), second-order central-difference scheme is used.

3.3.3. Projection method

Given $\mathbf{u}^n$ and $\mathbf{f}^n$, the second-order Adams–Bashforth scheme is used for time advancement and are written as follows:

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} + \left(\frac{3}{2}\mathbf{A}^n - \frac{1}{2}\mathbf{A}^{n-1}\right) + \frac{1}{\rho(\phi)}\frac{\delta \mathbf{x}}{\Delta \mathbf{x}}P^{n+\frac{1}{2}} = 0, \tag{45}$$

where $\delta\mathbf{x}$ is the first-order central-difference operator and $\mathbf{A}$ denotes

$$\mathbf{A} = \mathbf{u}\cdot\nabla\mathbf{u} - \frac{1}{\rho(\phi)}\nabla\cdot[\mu(\phi)(\nabla\mathbf{u} + (\nabla\mathbf{u})^T)] - \mathbf{g}, \tag{46}$$

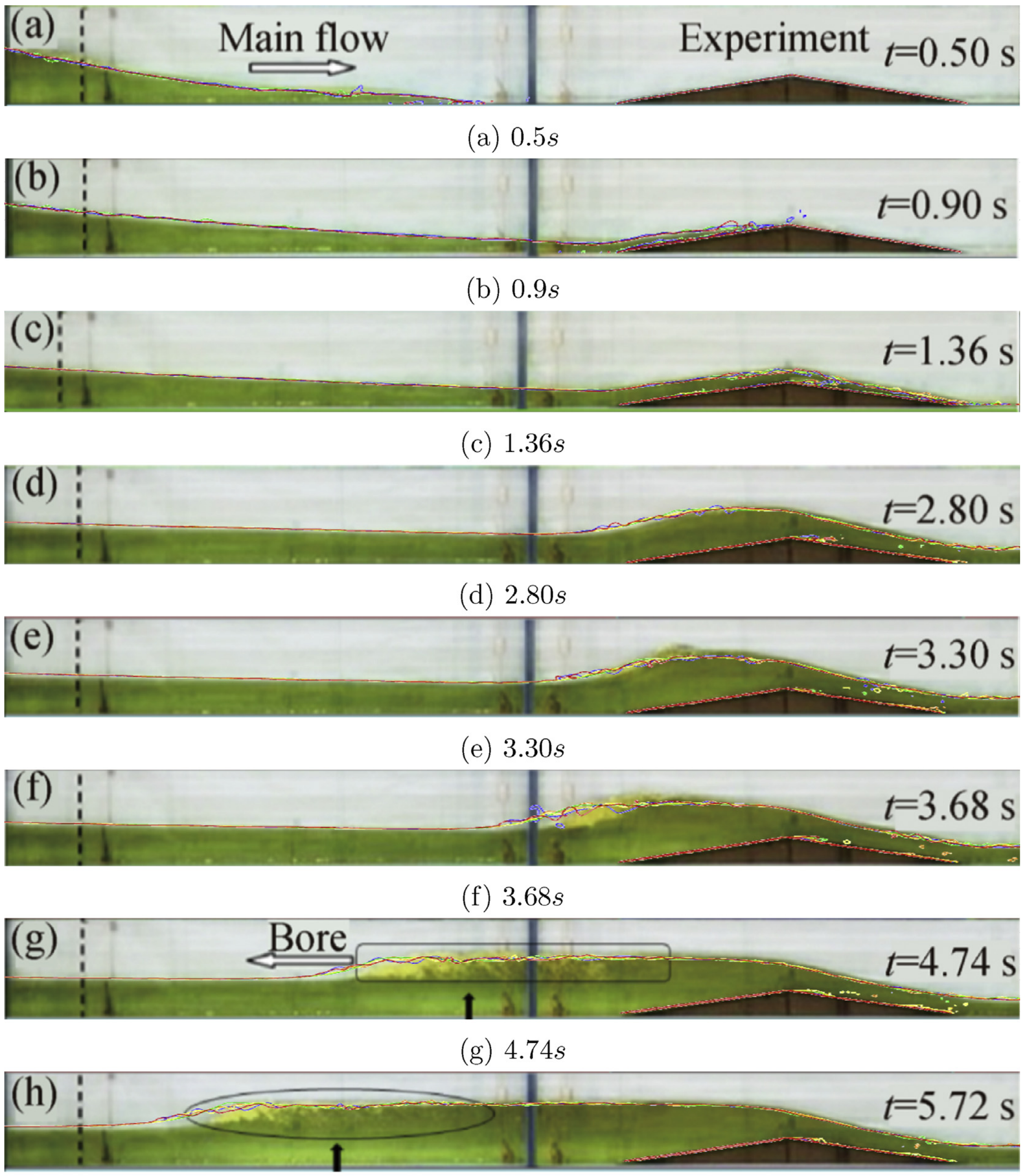
By using continuity equation and the divergence operate in Eq. (46), the Poisson equation for $P^{n+\frac{1}{2}}$ can be derived as

$$\left(\frac{\delta x}{\Delta x}\frac{1}{\rho(\phi)}\frac{\delta x}{\Delta x} + \frac{\delta y}{\Delta y}\frac{1}{\rho(\phi)}\frac{\delta y}{\Delta y} + \frac{\delta z}{\Delta z}\frac{1}{\rho(\phi)}\frac{\delta z}{\Delta z}\right)P^{n+1/2} = \mathbf{RHS}, \tag{47}$$

where $\mathbf{RHS} = \frac{\delta x}{\Delta x}\left(\frac{u^n}{\Delta t} - f^n\right) + \frac{\delta y}{\Delta y}\left(\frac{v^n}{\Delta t} - g^n\right) + \frac{\delta z}{\Delta z}\left(\frac{w^n}{\Delta t} - h^n\right)$. Eq. (47) is discretized as

$$aP_{i-1,j,k} + bP_{i+1,j,k} + cP_{i,j,k} + dP_{i,j-1,k} + eP_{i,j+1,k} + fP_{i,j,k-1} + gP_{i,j,k+1} = (\mathbf{RHS})_{i,j,k}. \tag{48}$$

A point successive over-relaxation method is used for the above tridiagonal and sparse matrix of pressure. The convergence tolerance $\varepsilon = 10^{-4}$ is used for the solutions calculated from two consecutive iterations $P^{m+1} - P^m$, where m is the iteration counter.



(a) 0.5s

(b) 0.9s

(c) 1.36s

(d) 2.80s

(e) 3.30s

(f) 3.68s

(g) 4.74s

(h) 5.72s

Fig. 4. Comparasion between experimental and numerical results of dam-break flow induced tsunami bore acting on the triangular breakwater.

4. Numerical validation

With the 3D parallel numerical solver developed by Fortran and OpenMP, first a typical problem which contains analytic solution was adopted to valid the numerical accuracy of IBM. Then, one dam-break induced tsunami bore problems with stationary triangular breakwater is numerically carried out for mesh convengence study, mass conversation analysis and wave elevation verification respectively.

4.1. Numerical validation for immersed boundary method (IBM)

To verify the accuracy of IBM, one problem which contains analytic solutions is investigated. The validation problem is solved in the a square domain with side length $l = 1$. Within the domain, there has a circular object centered at (0.5, 0.5) in $X - Y$ plane. The radius of the circular object is 0.2. The schematic in $X - Y$ view of this problem is shown in Fig. 1.

Boundary condition has an analytic expression as follow

$$u(x, y, z) = \frac{-2(1+y)}{(1+x)^2 + (1+y)^2}, \tag{49}$$

$$v(x, y, z) = \frac{2(1+x)}{(1+x)^2 + (1+y)^2}, \tag{50}$$

$$w(x, y, z) = 0; \tag{51}$$

After solve the N-S equation, the pressure field has exact solution as follow

$$p(x, y, z) = \frac{-2}{(1+x)^2 + (1+y)^2}. \tag{52}$$

We use L_2 norm to measure the numerical error with three different uniform cell size (0.05; 0.025; 0.0125). The numerical error results are shown in Table 2

It can be seen in Table 2 that the numerical errors that the predicted

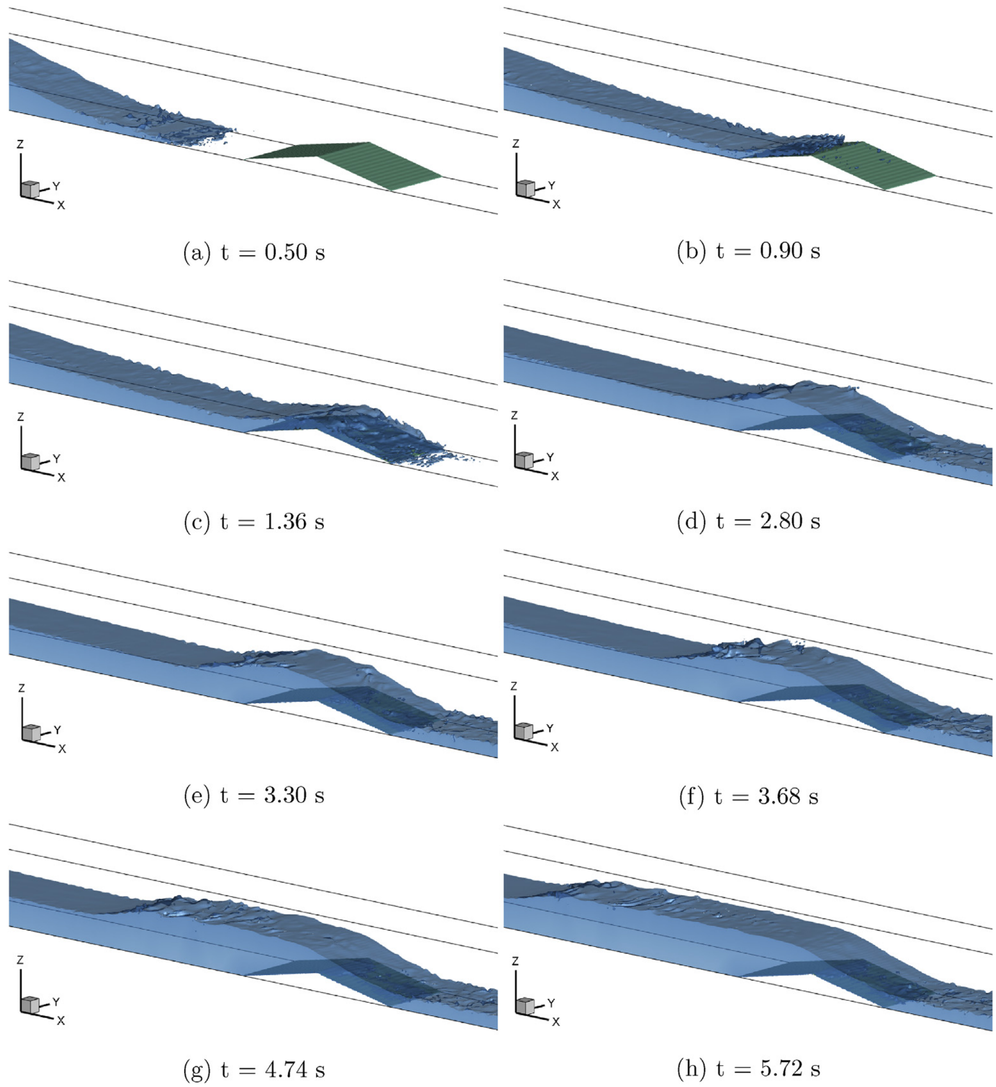


Fig. 5. Numerical free surface iso-surface of dam-break flow induced tsunami bore acting on triangular breakwater by CLSVOF and IBM with different time.

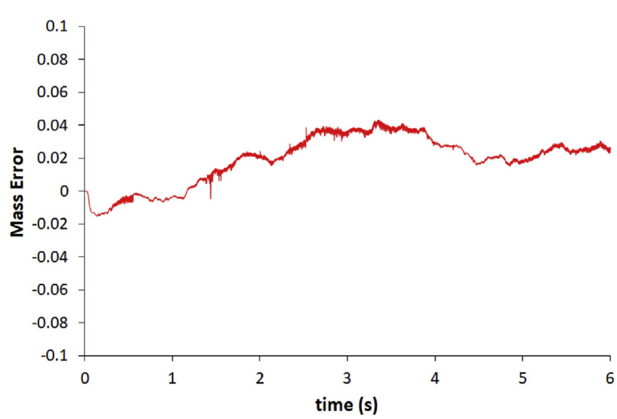


Fig. 6. The mass error $\mathbf{M}_{error}$ of tsunami bore impact on a stationary triangular breakwater by using CLSVOF with IBM.

solutions agree very well with the exact solutions. The global second order accuracy for velocity and pressure can be obtained.

4.2. Numerical validation for dam-break induced tsunami bore with stationary triangular breakwater

4.2.1. Parameters and setting for numerical simulations

We choose a model experiment as a benckmark case and for comparison. According to the referenced literature (Hatice et al., 2014) as shown in Fig. 2, the computational domain was 8.90 m long, 0.30 m wide and 0.30 m high for simulation. All the surfaces of the channel bottom and sidewalls were assumed as smooth. A volume of water in the reservoir 4.65 m long, 0.30 m wide and 0.25 m high was defined as an initial condition. According to the model experiment, the probes was located along the channel as shown in Fig. 2 ($P1 - P6$). The upstream boundarys was set as wall, since no flow entered into the reservoir, and downstream boundary was set as velocity outlet. The channel sidewalls were chosen as wall. Tangential and normal velocities were zero at the solid boundary due to defining the no-slip condition. The computational domain was subdivided into a mesh of fixed rectangular cells using Cartesian coordinates. The gate was rapidly opened in a very short time ($t = 0.06 - 0.08\ s$). The effect of triangular breakwater was simulated by IBM. Within the experiment, the six measurement probes of stage hydrographs are setted along the channel ($P1 - P6$), to obtain

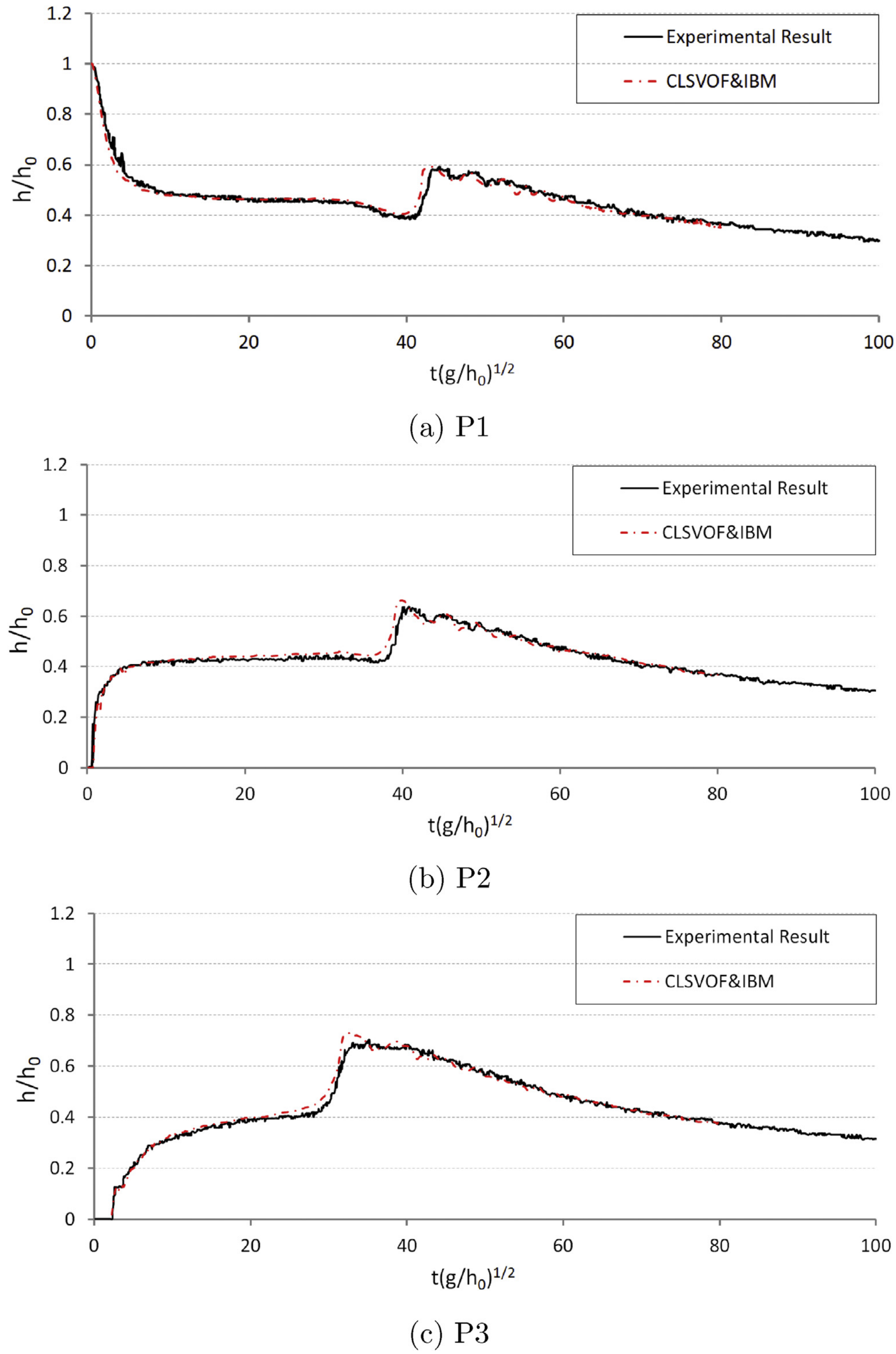
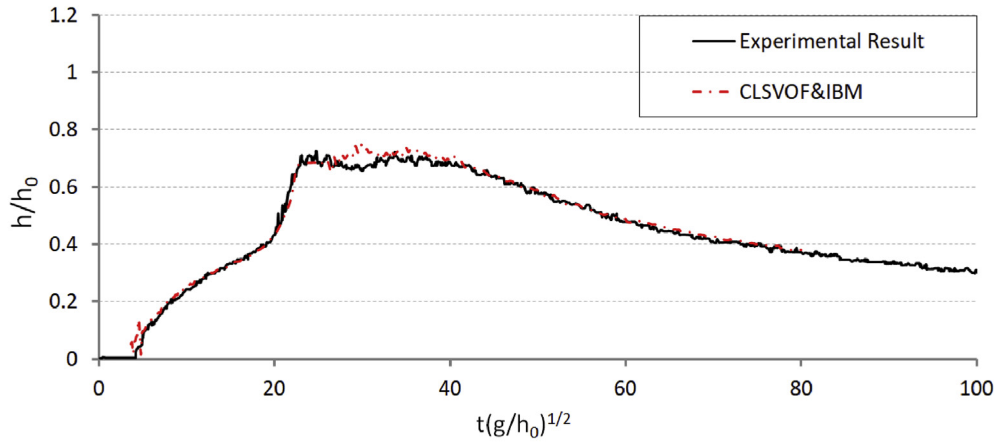


Fig. 7. Numerical and experimental results comparasion for P1-P3.

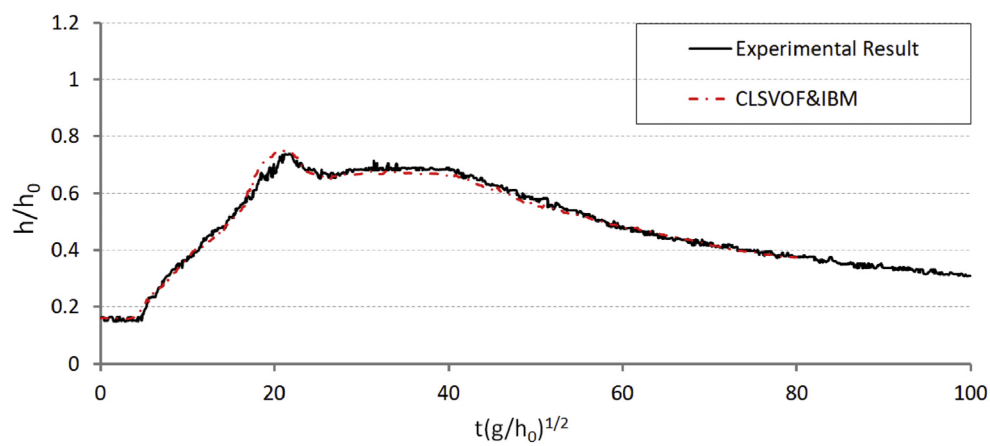
water elevation variations over time by digital video images at selected sections. The corresponded numerical probes are also setted in the same location during the numerical simulation.

4.2.2. Mesh convergence study

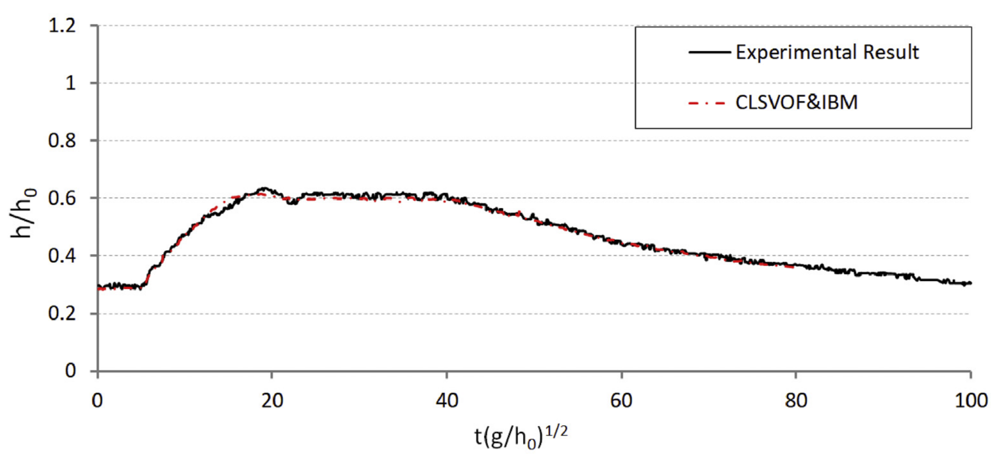
Fig. 3 shown the experimental water level for probe $P3$ and $P5$ compare with the temporal evolution calculated in grid interval $5mm$, $10mm$ and $20mm$ using CLSVOF and IBM. In Fig. 3, time t was multiplied by $\left(\frac{g}{h_0}\right)^{0.5}$ to get dimensionless time (T) where g is the acceleration of gravity. It is clear that the numerical results converge to the experimental results as the grid interval reduced from Δ= 20mm to Δ= 5mm. The water levels in finer grid interval $5mm$ quite well with the experiment result in Ref. (Hatice et al., 2014). After ensuring that the grids are independent, the grid interval $5mm$ (i.e. $\Delta x = \Delta y = \Delta z = 5mm$) is carried out in the other numerical cases in this research.



(a) P4



(b) P5



(c) P6

Fig. 8. Numerical and experimental results comparasion for P4-P6.

4.2.3. Numerical verfication on free surface evolution

Comparasion results of free surface evolution between numerical and experimental results are shown in Fig. 4, as well its corresponding 3D free surface behaviors at different times are shown in Fig. 5. Within Fig. 4, three lines with different section location along y axis were used to represents free surface deformation at each corresponding time. The red line represents section with $y = 0.05\ m$, blue line and yellow line represents $y = 0.15\ m$ and $y = 0.25\ m$ respectively.

In Fig. 4 and corresponding iso-surface profile within Fig. 5, it can be observed that when the gate was removed, a strong dam-break wave is generated which propagates over the dry channel. When the dam-break induced tsunami bore arrives at the breakwater, a part of the wave is reflected and a spilling type reflected bore appears travelling along the upstream direction, while the other part travels along the

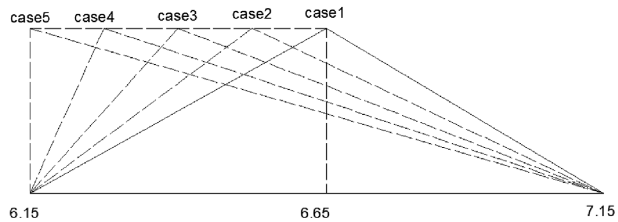


Fig. 9. The schematic diagram of different cases (unit: m).

downstream of the channel. At $t = 0.90\,s$, the tsunami bore arrives at the top of the breakwater (Fig. 4a). At $t = 1.36\,s$, the tsunami bore just flows past the right toe of the breakwater (Fig. 4b). In this period the numerical water front agrees well with the experiment images qualitatively. At $t = 2.80\,s$, the spilling type reflected bore begins to emerge (Fig. 4d). After $t = 3.30\,s$, the spilling type reflected bore begins to move in the upstream direction (Fig. 4e–h). The spilling type wave breaking is observed in both experimental snapshots and the present computed results after $t = 3.30\,s$. It can be observed that the numerical solver presents good accordance with experimental snapshots in the spilling type reflected bore and wave breaking. However, slight discrepancy exists in the propagation speed between the experiment observations and numerical predictions. The experiment bore has a little time delay. With the assumption of frictionless solid boundaries, the propagation of the spilling type reflected bore is over-predicted under the frictionless condition in the present IBM model. The time delay caused by the gate motion in the experiment also could causes such discrepancy.

It can be found from Figs. 4 and 5 that the numerical solver based on CLSVOF and IBM gaves very good agreement with experimental result. CLSVOF method can handle such strong nonlinear phenomenon very well with enough numerical accuracy.

4.2.4. Mass conservation analysis

Beside of numerical verfication on free surface evolution, the ability of mass conservation was also carried out for tsunami bore impact on a stationary triangular breakwater problem. The mass error M_{error} is defined as follow

$$M_{error} = \frac{|M_t - M_0|}{M_0} \times 100\%, \tag{53}$$

where $M_t (= \int_\Omega C(t)\, d\Omega)$ is the mass at the time during the computation and $M_0 (= \int_\Omega C(0)\, d\Omega)$ is the mass at the beginning of the computation. The mass error of tsunami bore impact on a stationary triangular breakwater by using CLSVOF and IBM during the simulation is presented in Fig. 6. It can be seen that using CLSVOF gives a good mass

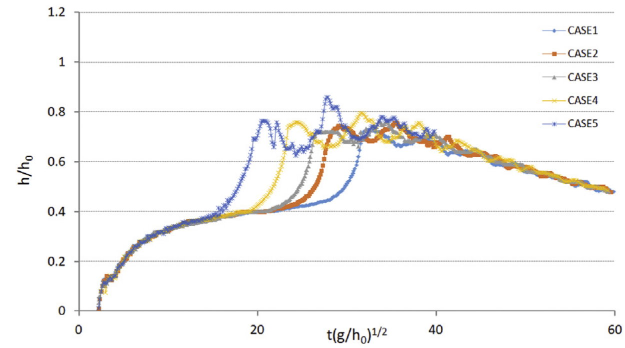


Fig. 11. Wave elevation probed time history curves at *P*3 with different slopes.

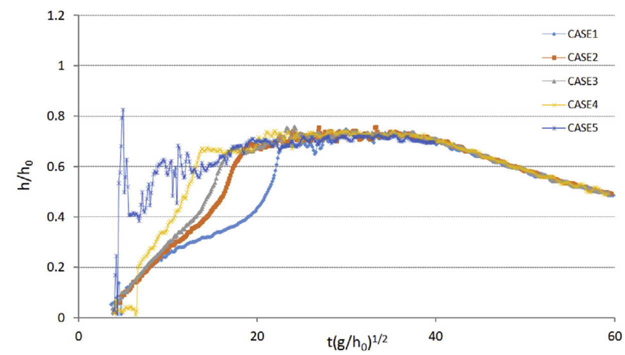


Fig. 12. Wave elevation probed time history curves at *P*4 with different slopes.

conservation result, which result from the CLSVOF is transported using conservative equations which reduce mass conservation errors while retaining the accuracy of interface normal vector even in break-up phenomena. In other words, the CLSVOF which combines the advantages of THINC/WLIC and LS schemes makes the nonlinear phenomena of free surface approaching to the experimental results.

4.2.5. Numerical verfication on free surface elevation along the channel for different probes

Fig. 7 and Fig. 8 shows the comparison between the numerical and experimental time history curves of water depths at six probes from P1 to P6. Agreement between the numerical and experimental results is well achieved. Probe P1 (Fig. 7a) and Probe P2 (Fig. 7b) are located just a little distance from the gate along upstream and downstream respectively. When the gate is removed, a strong dam-break induced tsunami wave is generated, then propagates along downstream,

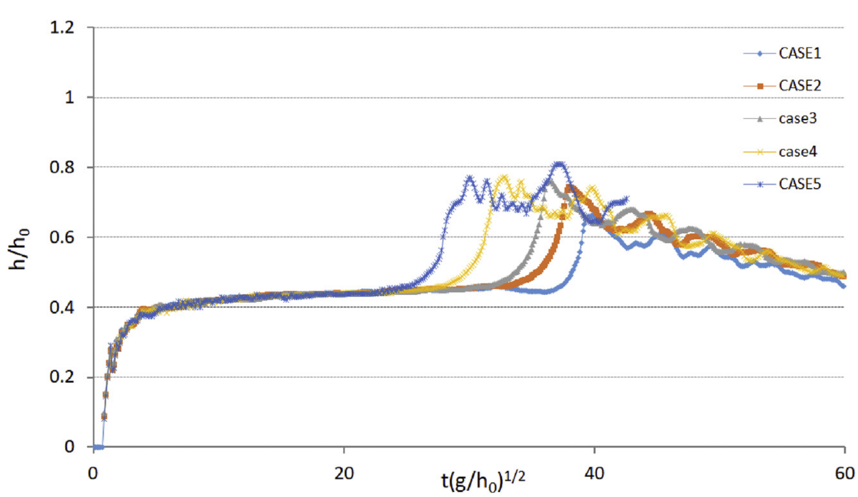


Fig. 10. Wave elevation probed time history curves at *P*2 with different slopes.

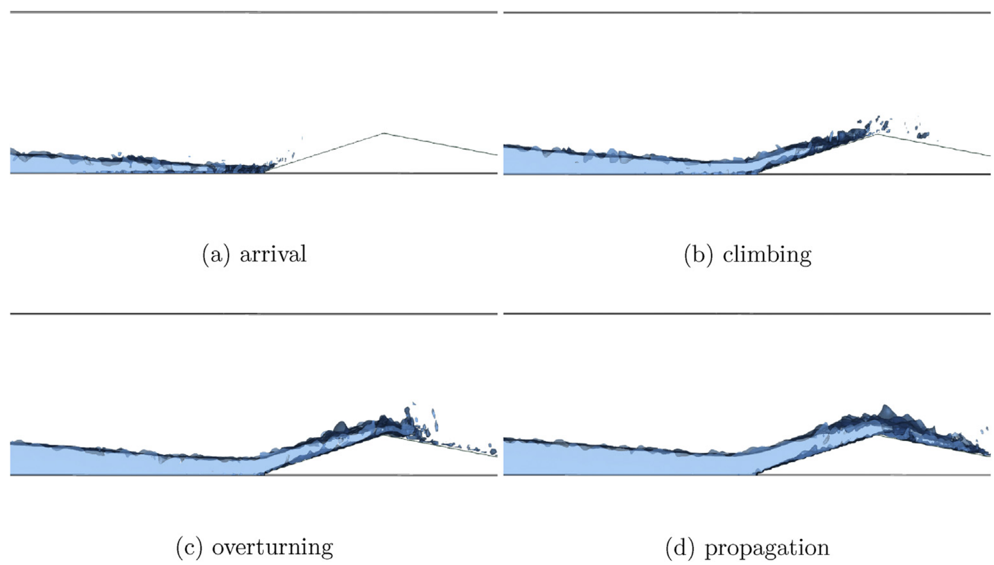


Fig. 13. Numerical free surface iso-surface of dam-break flow induced tsunami bore acting on triangular breakwater by CLSVOF and IBM with different stage of case 2.

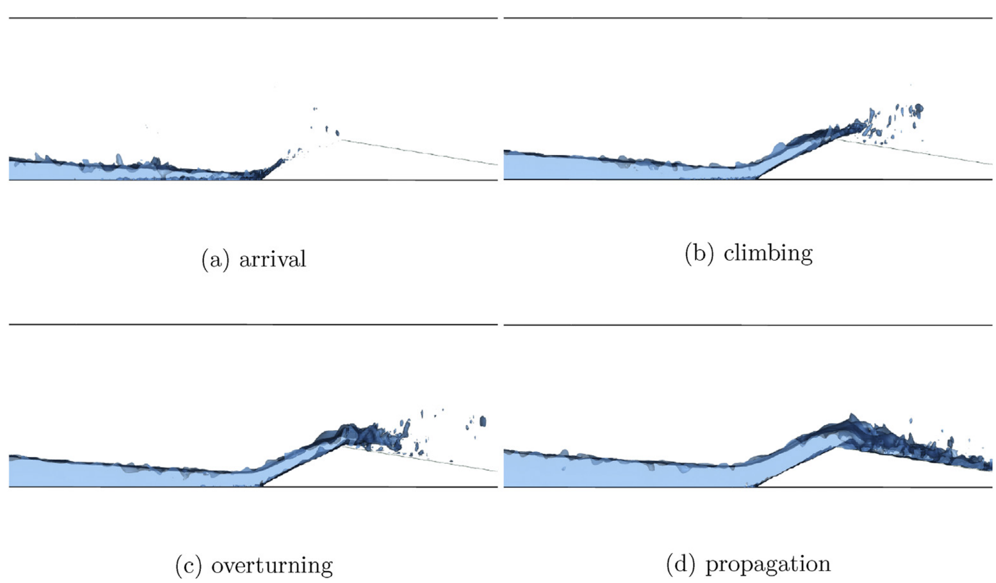


Fig. 14. Numerical free surface iso-surface of dam-break flow induced tsunami bore acting on triangular breakwater by CLSVOF and IBM with different stage of case 3.

meantime the abrupt rise in water depth are observed at P1 and P2 in both numerical results and experimental data ($t^* < 5$). There just has a little change in water depth at P1 and P2 during ($t^* = 5 - 40$). An abrupt rise in water level at about $t^* = 40$ as the negative bore from the breakwater can be observed in both the numerical and experimental results. After the rising the water depth gradually drop due to the reduction of the incoming flow. From Fig. 7a and b, the present model slightly overestimates the water levels in $t^* = 40 - 42$ in Fig. 7a and $t^* = 35 - 40$ in Fig. 7b where the negative bore just reaches the water level measurement positions. The possible reason for this difference is that water levels were obtained directly from the digital video images, that could ignoring the splash in front of the negative bore in the experiment. In Fig. 7c for probe P3, a rising of the water level during $t^* = 2 - 30$ can be observed, and an abrupt rising resulted from the arrival of the negative bore is observed at $t^* = 30$. The results at $t^* = 30$ of the present model for P3 reveal similar phenomena to those at P1 and P2. Some overestimates can be seen during $t^* = 32 - 40$ within P3, that may be due to the spilling type wave breaking in front of the negative bore, and the splash of the free surface. In Fig. 8a for probe P4, compare with the experimental result, the numerical result shows rapid rising in water level, during the negative bore through the measurement location of P4. Such phenomena may be resulted from the splashes of the wave breaking in front of the negative bore, and the hydraulic jump on the free surface are ignored in the experimental measurements. In Fig. 8b for probe P5, it can be observed that there has nearly no elapsed time for travelling between flood positive flow and negative flow, since the rapid rising of the water level occurs behind the negative bore. Therefore, water levels constantly rise up to their maximum levels. At the measurement locations close to the toe of the breakwater (P4, P5 and P6), the maximum water level occurs.

By comparison, it can be found that the numerical method of this paper can better reproduce the quantitative changes of different liquid levels for each section over time. Numerical result of water level of all probes (P1-P6) are always in good agreement with the experimental data during the simulation. It also can be observed that the numerical approaching adopted in this paper has very good ability and accuracy

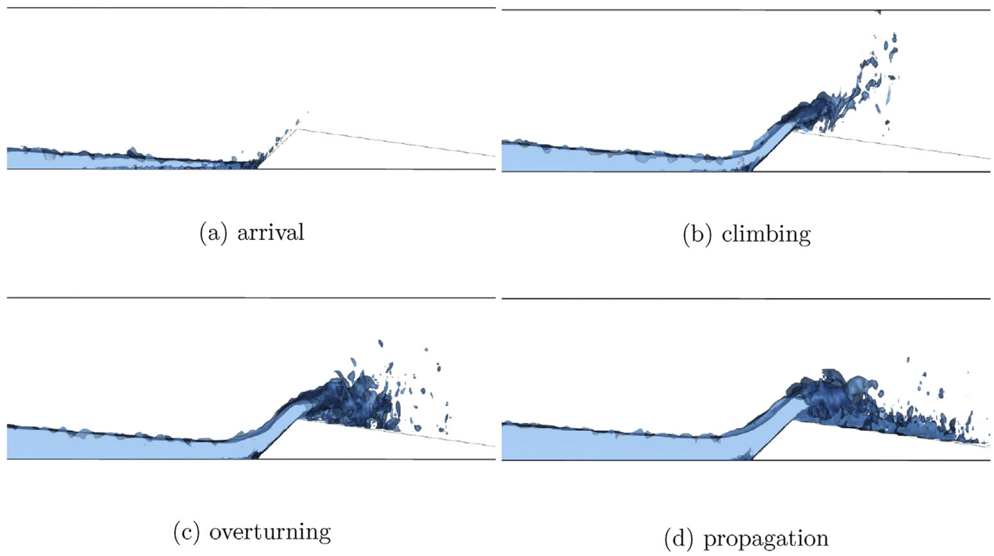


Fig. 15. Numerical free surface iso-surface of dam-break flow induced tsunami bore acting on triangular breakwater by CLSVOF and IBM with different stage of case 4.

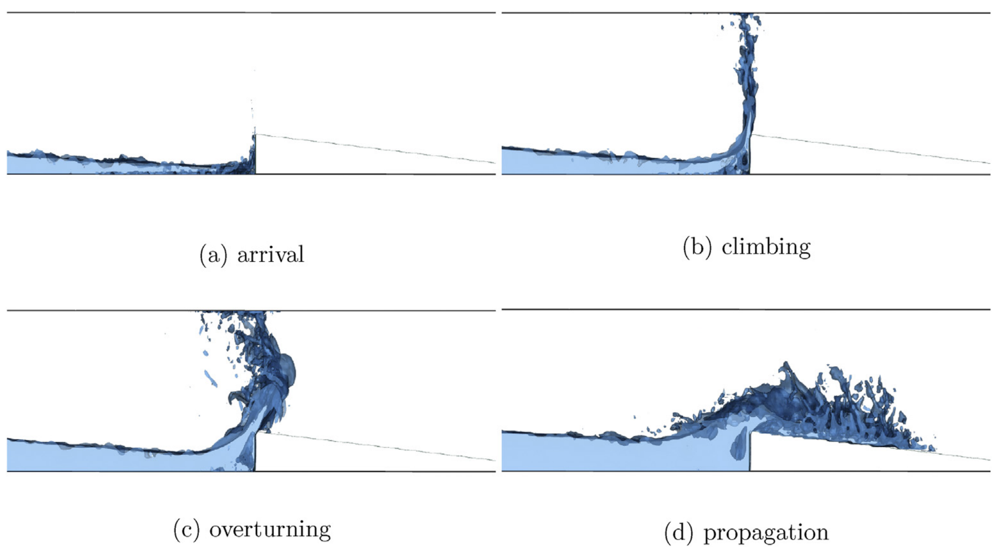


Fig. 16. Numerical free surface iso-surface of dam-break flow induced tsunami bore acting on triangular breakwater by CLSVOF and IBM with different stage of case 5.

on wave elevation capturing.

5. Numerical results for tsunami bore acting on breakwater with different slopes

In this chapter, we will investigate the interaction between tsunami waves and breakwaters with different slopes. Based on the previous chapter, we added four numerical cases with different slopes. The schematic diagram of different cases is shown in Fig. 9. As can be seen from Fig. 9, case 1 is the benchmark case that was adopted in the previous chapter for numerical verification. The slope of the breakwater increases gradually from the case 2 to the case 5. In numerical simulation, the setting of numerical parameters is basically the same as that in the previous chapter. The difference lies in the slope of the breakwater and the length of the numerical flume. The length of the numerical flume is re-set as 12 m, in order to investigate the energy dissipation of the tsunami wave propagating along downstream after acting with the breakwater. The position of the left toe of the breakwater keeps unchange among five cases, so the kinetic energy of the tsunami wave is the same when it touches the left toe of breakwaters.

Figs. 10–12 are the water level elevation time history curves for P2 to P4, respectively. As can be seen from Figs. 10 and 11, as the slope increases gradually, the arrival time of the reflected wave to the measuring probes P2 and P3 is also gradually advanced. However, the maximum water level rising of the reflected waves for five different slopes does not change too much, and gradually tends to the same value with time. However, the nonlinear phenomenon of the reflected wave was increases with the increasing of the slope. After the first rise of the water level, the change of the water level with time becomes more and more irregular, with the increase of the slope. As can be seen from Fig. 12, the water level of case 5 with vertical slope varies dramatically, which means a strong breaking wave phenomenon occurs here.

Figs. 13–16 is a schematic diagram of the free surface deformation at four typical moments, i.e. the arrival, climbing, overturning, and propagation of the interaction between tsunami wave and breakwater for case 2–5, respectively. As can be seen from Fig. 13, the slope of case 2 is a little higher than that of case 1, but there is no obvious nonlinear phenomenon occurs. The water body is relatively uniform, and only in the process of climbing and overturning, there has slight breaking waves phenomena at the wave head. As can be seen from Fig. 14, with the further increase of the slope, in the climbing process, the wave head appeared more obvious nonlinear phenomenon, and in the process of overturning, the wavefront appeared splash phenomenon, while in the propagation process of nonlinear phenomenon became relatively strong. As the slope further increases, it can be seen in Fig. 15 that nonlinear phenomena such as splash and wave breaking of the wavefront are severe, and the wavefront is almost broken. In Fig. 16, we can see that the nonlinear phenomena are very serious, the water body is almost completely broken in the process of overturning, and the three-dimensional effect is very strong, meantime the energy dissipation caused by strong turbulence can be created here. The results show that the CLSVOF-THINC/WLIC-IBM method developed in this paper can well simulate the strong nonlinear phenomena and time evolution of the interaction between tsunami waves and breakwater.

In Table 2, the listed time are wavefront reaches the position of left end of breakwater where located at 6.15 m, and the position of 10 m and 12 m of the water channel respectively. From the result of the table, we can see that the velocity of wavefront of case 5 is the slowest. Because the kinetic energy of the tsunami wave keeps the same when it touches the left toe of breakwater within five cases, it can be seen that the dissipation of kinetic energy occurs after the tsunami wave acting with the breakwater, and such kinetic energy dissipation increases with the increase of the slope of the breakwater. It can also be seen from the table that from case 1 to case 3, the time of wavefront arriving at the right end of the water channel has no too much difference. It can be observed that strong nonlinear phenomena such as turbulence, wave breaking, splash are the biggest source of kinetic energy loss, during the interaction between tsunami wave and the breakwater.

6. Concluding remarks

In this study, the interaction between dam-break induced tsunami waves and triangular breakwater was numerically investigated over initially dry downstream channel with five distinct slopes, by means of combined 3D CLSVOF-THINC/WLIC-IBM numerical approaching. Through the process of tsunami waves over the triangular breakwater, we have compared the numerical simulation results of water elevation at different sections around the breakwater over time with experimental data. The comparison shows that the numerical model we developed in this paper not only can better reflects the entire process of the dam-break induced tsunami wave over the breakwater, but also is able to reproduces the rich physical phenomena during the process, such as changes in wave surface curvature, jets, wave breaking, and other strong nonlinear free surface large deformation phenomena. The findings of detailed numerical research show that the impact of dam-break induced tsunami wave on breakwater causes wave reflection, when the tsunami wave cross after the breakwater, the wave speed was turn to slow resulted from energy dissipation due to the existed breakwater. Strong nonlinear phenomena such as turbulence, wave break, splash are the biggest source of kinetic energy dissipation, during the interaction between tsunami wave and the breakwater. To study the interaction between tsunami waves and breakwater, the numerical model adopted in this paper is practical and reliable. Additionally, the results of this study can also provide reliable reference for the future design of coastal and harbor tsunami phenomenon prevention.

Acknowledgements

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